🔗 Topic 2 at a glance: how the three papers connect

- **Function adapts to form** more than form follows function: the tongue compensates for malocclusion in speech (Johnson and Sandy) and in swallowing (Proffit and Mason).
- **Duration is what moves teeth:** sucking habits lasting years change the arches (Warren and Bishara), and the resting tongue matters, while a brief swallow or a spoken sound does not.
- **Correlation is not causation:** speech defects, tongue thrust and bottle-feeding all co-exist with malocclusion without proving they cause it.
- **Clinically:** treat teeth for teeth and speech with a speech therapist; stop pacifier and digit habits by about 2–3 years; do not treat tongue thrust before puberty.

Tooth Position and Speech: Is There a Relationship?

by **Johnson and Sandy** · Angle Orthod 1999;69(4):306–310 · Review article, University of Bristol

📚 Study purpose

To review whether **tooth position (malocclusion) affects speech**, and whether **changing tooth position improves speech**, and to explain why this is so hard to study.

🗣️ Background

- Speech uses 3 systems: **respiratory, phonatory, articulatory**. Teeth, lips and tongue articulate consonants by blocking or shaping airflow.
- Articulatory disorders make up **50–60 %** of all speech disorders.
- **Harvold's 3 possible links:** 1. coincidence; 2. a common cause (genetic or metabolic CNS disorder giving poor motor control *and* distorted morphology); 3. true cause and effect.

🔍 Evidence by occlusal trait

Trait	Key findings	Verdict
Malocclusion in general	Fymbo (410 students): more trouble with dental sounds; Hopkin & McEwen (101 speech-clinic children): defects as likely with normal occlusion; Oliver & Evans (35): poor articulators had *more normal* teeth	No consistent link
Anterior open bite	Fymbo: **63 %** defective speech, **4 %** superior; Bernstein (437 + matched controls): only open bite strongly linked, to **lisping**, but lisp severity ≠ open-bite size; Laine: consonants made too far forward **×3.4**	**Most consistent link**; alone usually mild, worse with Class III
Class III (mesial occlusion)	Low, flaccid tongue; Guay (12 anterior crossbites): normal /s/ in only 1; Laine: **×4.5** mesial occlusion, **×3.7** mandibular overjet; Bloomer: mild cases speak normally	Link; worse when combined with open bite
Increased overjet	Tongue adapts (Subtelny, Jensen); Blyth (200): skeleton had little effect; Laine: /s/ distortion	Link to /s/
Deep overbite	Lubit (300): /s/ link but negligible overall; Ingervall & Sarnas: lateral lisp; Laine: none	Conflicting
Lateral crossbite, narrow palate	Laine: **×1.7**; others dispute	Weak
Upper spacing, missing upper incisors	Laine: /l/ /n/ /d/ /r/; many compensate; none for lower spacing or crowding	Weak

🦷 Does treatment change speech?

- **Rathbone & Snidecor:** 8 patients, 4 years of orthodontics, no speech therapy: faulty sounds **6.4 → 1.5**; /s/ /z/ /sh/ /zh/ stayed faulty.
- **RCT, 25 patients with functional voice disturbances:** occlusal adjustment improved voice at 2 months; placebo unchanged.
- **Orthognathic surgery:** Higley (2 cases) mixed; Goodstein (5) no change; **Dalston & Vig (40 women) no change**; Ruscello's review and Yamaguchi (15 Class III, no statistics) improved; errors may *increase* after surgery. Evidence equivocal: no guarantees.

⚠️ Why the research is hard

3 study types: speech patients checked for malocclusion, malocclusion patients checked for speech, unselected populations (the only nonselected type, but many done on undergraduate students). Definitions vary, reliability is rarely reported, languages differ, and there is no animal model.

🧠 Main conclusions

1. Most consistent traits: **Class III, anterior open bite, increased overjet, spacing**.
2. Severity of malocclusion does not predict severity of misarticulation (Fymbo saw a link; Rathbone and Bernstein did not).
3. Sounds learned last in childhood are the ones most often distorted.
4. **Adaptation decides (Bloomer):** normal structure + normal function = normal speech; abnormal structure + adaptive function = normal; abnormal structure + no adaptation = abnormal; normal structure + abnormal function = abnormal.
5. No substantial evidence that orthodontics corrects articulation disorders: **refer to a speech therapist**.

✅ Number	Meaning
50–60 %	articulatory share of all speech disorders
63 % vs 4 %	open bite: defective vs superior speech (Fymbo)
×4.5 / ×3.7 / ×3.4 / ×1.7	risk of consonants made too far forward: mesial occlusion / mandibular overjet / open bite / lateral crossbite (Laine)
6.4 → 1.5	faulty sounds before and after orthodontics, 8 patients

Duration of Nutritive and Nonnutritive Sucking Behaviors and Their Effects on the Dental Arches in the Primary Dentition

by **Warren and Bishara** · Am J Orthod Dentofacial Orthop 2002;121(4):347–356 · University of Iowa

📚 Study purpose

To find whether **how long** a child breast-feeds (nutritive sucking) or uses a **pacifier or digit** (nonnutritive sucking) changes the dental arches and occlusion at the **end of the primary dentition**.

🔬 Study design

- **Prospective birth cohort** (Iowa, mothers recruited 1992–1995, about 700 children). Parent questionnaires at 3, 6, 9, 12, 16 and 20 months; sucking-habit questions continued at 24 months, then yearly.
- Casts at **4.5–5 years**: 547 children; 98 excluded (a permanent tooth present), 56 excluded (missing habit data): **372 analysed** (as printed; 547 − 98 − 56 = 393).
- Grouped by habit type (pacifier or digit; 45 unclassifiable) and duration: **<12, 12–24, 24–36, 36–48, ≥48 months**. Children with <12 months of nonnutritive sucking (n = 119) were grouped by breast-feeding: none, <6, 6–12, >12 months.
- Measured arch widths, lengths and depths, palatal depth, overjet, overbite, posterior crossbite, open bite and canine class; calipers to 0.05 mm, two examiners. ANOVA, chi-square and t-tests, P < .05.

📊 Key results

Breast-feeding: **no effect** on any arch measurement or occlusal trait (every P ≥ .114).

Habit and outcome	<12 mo	24–36	36–48	≥48	P
Pacifier: posterior crossbite %	6.6	16.3	22.2	**41.7**	.001
Pacifier: anterior open bite %	5.5	0	0	**25.0**	.008
Pacifier: mandibular canine width, mm	22.6	22.5	**23.6**	23.4	.041
Pacifier: overbite, mm	1.9	2.0	1.7	**0.4**	.007
Pacifier: palatal depth, mm	14.8	14.3	13.9	**13.3**	.001
Digit: anterior open bite %	5.5	0	15.4	**37.0**	.001
Digit: overjet ≥4 mm %	6.6	8.7	15.4	**39.1**	.001
Digit: overjet, mm	2.4	2.2	3.2	**3.7**	.001
Digit: maxillary canine width, mm	28.8	29.0	29.0	**27.8**	.009
Digit: maxillary canine depth, mm	10.0	10.0	10.8	**11.2**	.001
Digit: posterior crossbite %	6.6	0	0	15.2	.099 NS

Tables III–VI; the 12–24-month column is omitted for space. **Not affected:** Class II canine relationship (pacifier P = .210; digit P = .161) and overjet with a pacifier (P = .683).

	🏷️ Pacifier	👍 Digit (thumb or finger)
Signature	**Posterior crossbite**: lower arch wider, upper slightly narrower	**Overjet**: upper arch narrower and deeper
Open bite	Yes, symmetrical	Yes, often **asymmetrical**, on the side of the hand used
Starts to matter	Beyond **24 months**	**36–48 months**
Head to head	More crossbite (P = .034, .044); wider lower arch (P = .013)	More overjet (P = .001); more overjet ≥4 mm (P = .012)

🧠 Main conclusions

1. Breast-feeding length made no difference (in children with little nonnutritive sucking).
2. Both habits are linked to **open bite and reduced overbite**; **pacifier → posterior crossbite**; **digit → overjet with a narrow, deep upper arch**.
3. Some changes persisted well after the habit stopped: pacifiers stopped by 36 months still gave 16.3 % crossbite at age 5; digits stopped by 48 months gave 15.4 % open bite and 3.2 mm overjet.
4. Advice to worry only if a habit lasts into the early mixed dentition (AAO, ADA, AAP, AAPD) may be too late: habits beyond **2–3 years** clearly raise the risk.

⚠️ Limitations

Iowa only, few nonwhite children; habits reported by parents and not verified; small subgroups (breast-feeding >12 months n = 13; pacifier ≥48 months n = 12); primary dentition only, so persistence into the mixed dentition is unknown.

✅ Number	Meaning
372	children analysed at 4.5–5 years
41.7 % vs 6.6 %	posterior crossbite: pacifier ≥48 months vs <12 months
37.0 % / 39.1 %	open bite / overjet ≥4 mm with digit habits ≥48 months
66.7 % / 71.7 %	at least one problem trait: pacifier / digit ≥48 months
2–3 years	age after which habits clearly raise malocclusion risk

Myofunctional Therapy for Tongue-Thrusting: Background and Recommendations

by **Proffit and Mason** · J Am Dent Assoc 1975;90(2):403–411 · Review with the authors' pressure research, Florida and Kentucky

📚 Purpose

To review how **form and function interact** in tongue thrust and give **guidelines for selecting patients for myofunctional (swallowing) therapy**.

👅 What is tongue thrust? Three forms

Swallow: tongue tip forward between the incisors, touching the lower lip. **Speech:** tongue fronted against or between the incisors, jaw open. **Rest:** tongue forward, jaw slightly open. All three occur in people **with and without** speech or dental problems.

🔄 Habit or normal development?

- **Infant swallow:** jaws apart, tongue between the gum pads touching the lower lip, active lips. **Adult swallow:** teeth together, relaxed lips, tongue tip on the palate behind the incisors.
- The adult pattern appears **between 2 and 12 years**; many normal children finish only near puberty. Tongue thrust is usually a delayed normal stage, not a learned habit.
- Straub blamed bottle-feeding (478 patients, 2 breast-fed): correlation, not cause. Thumb-sucking delays the transition.

🫁 Why young children carry the tongue forward

- The tongue grows like neural tissue (near full size by about 8 years); the mandible grows later, so the tongue is relatively big.
- Tonsils and adenoids peak before puberty: the tongue comes forward to keep the airway open; nasal obstruction leads to mouth breathing.
- At puberty the ramus grows, lymphoid tissue shrinks and the cervical vertebrae grow taller: more room, so the tongue moves back.

📊 Key evidence

Finding	Number
Anterior open bite, US children 6–11 years (USPHS, about 8,000)	**5.7 %** (black 16.3 %, white 3.9 %)
Open bite ≥2 mm	black 9.6 %, white 1.4 %
Tongue-thrust swallow	about **50 %** at 6 years, **25 %** at 15
Thumb-sucking (USPHS)	about 10 %
Simple open bites that self-correct (1,400 Navajo children)	**80 %**
Excess maxillary incisor protrusion	about 17 %: more than thumb-sucking, so not mainly thumb-sucking; genetics a significant cause
Myofunctional therapy, controlled studies (Subtelny)	does **not** significantly change the swallow pattern

Tongue thrust and thumb-sucking are far more common than the open bite they are said to cause.

⚖️ Pressure studies: the equilibrium question

- Tongue and lip pressures during swallowing are **not balanced** and do not shape the arches: larger arches (Australian aborigines) show **lower** tongue pressure; protruded incisors get **less** tongue pressure, which returns to normal after retraction.
- Vertical tongue forces are **lower** in open-bite patients (Wallen): the opposite of the swallowing theory.
- **Resting posture matters more than swallowing:** eruption force is only about **5 g**, so a tongue resting forward may stop incisors erupting.

🧠 Recommendations

Situation	What to do
Tongue thrust only	No therapy; it has no preventive role
With a speech problem	Speech therapist gives articulation therapy now, never delayed for orthodontics; no swallowing therapy before puberty
With malocclusion	Treat the malocclusion first: the thrust disappears in about **80 %** (clinical experience)
Still present at puberty	Therapy for the **resting** tongue position, **combined with orthodontics**
Small airway (reduced faucial isthmus)	No swallowing exercises; tonsil removal decided by a physician on medical grounds

✅ Number	Meaning
2–12 years	when the adult swallow develops
up to 80 %	children with tongue thrust and open bite at 8 who improve by 12 without therapy